Numerical Solution of the Time-Dependent Schrödinger Equation via the Crank–Nicolson Scheme

Quantum tunnelling of a Gaussian wave packet
through a rectangular potential barrier

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TDSE_CN — theoretical notes

Atomic units throughout: $\hbar = m_e = e = 1$

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1 Introduction

The time-dependent Schrödinger equation (TDSE) governs the evolution of any non-relativistic quantum system. Even in one spatial dimension, it exhibits phenomena — wave packet spreading, interference, and quantum tunnelling — that have no classical analogue and are central to modern physics and technology (tunnel diodes, scanning tunnelling microscopy, nuclear alpha decay, ...).

Despite its conceptual depth, the one-dimensional TDSE with a static potential is fully tractable numerically: the problem reduces to propagating a complex vector in time via a sequence of sparse linear systems. This makes it an ideal testbed for implicit finite-difference methods and for developing intuition about quantum dynamics.

These notes describe the physical setup, the mathematical structure of the Crank–Nicolson discretisation, and the Thomas algorithm used to solve the resulting tridiagonal system at each time step. A final section analyses the output of the numerical simulation qualitatively and quantitatively.

2 Physical setup

2.1 The time-dependent Schrödinger equation

The state of a one-dimensional quantum particle is fully described by a complex-valued wave function $\psi(x,t) \in L^2(\mathbb{R})$. Its time evolution is governed by the TDSE

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} \;=\; \hat{H}\,\psi(x,t), \tag{1}$$

where the Hamiltonian operator $\hat{H} = \hat{K} + \hat{V}$ splits into a kinetic part

$$\hat{K} \;=\; -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \tag{2}$$

and a potential energy operator $\hat{V} = V(x)$ acting multiplicatively.

In **atomic units** ($\hbar = m_e = 1$) — used throughout this document — equation (1) becomes

$$\mathrm{i}\,\frac{\partial \psi}{\partial t} \;=\; \left[-\frac{1}{2}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi. \tag{3}$$

2.2 Physical interpretation of the wave function

The wave function $\psi(x,t)$ is not directly observable. The physically meaningful quantity is the *probability density*

$$\rho(x,t) \;=\; |\psi(x,t)|^2, \tag{4}$$

which gives the probability of finding the particle between x and $x + dx$ as $\rho(x,t)\,dx$. The total probability is conserved:

$$\int_{-\infty}^{+\infty} |\psi(x,t)|^2\,dx \;=\; 1 \quad \forall\, t. \tag{5}$$

This constraint — the *norm conservation* or *unitarity* of the evolution — is one of the most important consistency checks for any numerical scheme.

2.3 The Gaussian wave packet

The initial state chosen for this simulation is a *Gaussian wave packet*, the quantum state that most closely mimics a classical particle with a definite position and momentum:

$$\psi(x,0) \;=\; A\,\exp\left[-\frac{(x-x_0)^2}{4\sigma^2}\right]\exp(\mathrm{i}k_0 x)\,, \tag{6}$$

where:

- x_0 is the initial position of the packet centre,
- σ is the spatial width (standard deviation of ρ),
- k_0 is the central wave vector, corresponding to mean momentum $\langle p\rangle = \hbar k_0$,
- $A = (2\pi\sigma^2)^{-1/4}$ is the normalisation constant.

The Gaussian is the *minimum-uncertainty state*: it saturates the Heisenberg uncertainty relation $\Delta x\,\Delta p \geq \hbar/2$. Its Fourier transform is also a Gaussian, centred at k_0 with width $1/(2\sigma)$, so the momentum spread is $\Delta p = \hbar/(2\sigma)$.

The mean kinetic energy of the packet is

$$E \;=\; \frac{\langle p\rangle^2}{2m} \;=\; \frac{k_0^2}{2} \quad \text{(atomic units)}. \tag{7}$$

This is the quantity to compare against the barrier height V_0 in order to classify the dynamical regime.

2.4 The rectangular potential barrier

The external potential is a rectangular barrier of height V_0 and width w located at $x \in [x_{\mathrm{bar}},\, x_{\mathrm{bar}} + w]$:

$$V(x) \;=\; \begin{cases} V_0 & \text{if } x_{\mathrm{bar}} < x < x_{\mathrm{bar}} + w, \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

The barrier is the simplest non-trivial scattering potential and admits an exact analytic solution for plane-wave states (see Section 3), making it ideal for validating the numerics.

3 Quantum tunnelling

3.1 Stationary analysis

For a monochromatic plane wave $\psi(x) = \mathrm{e}^{\mathrm{i}kx}$ with energy $E = k^2/2$, the Schrödinger equation can be solved piecewise in each region. Matching wave functions and their derivatives at the barrier edges yields the exact transmission coefficient [1]

$$T \;=\; \left[1 + \frac{V_0^2 \sinh^2(\kappa w)}{4E(V_0 - E)}\right]^{-1} \qquad (E < V_0), \tag{9}$$

where $\kappa = \sqrt{2(V_0 - E)}$ (atomic units) is the evanescent decay constant inside the barrier. The reflection coefficient is simply $R = 1 - T$.

Several qualitative features are immediately apparent from (9):

1. $T > 0$ even when $E < V_0$ — this is *quantum tunnelling*. Classically, the particle cannot penetrate a region where $E < V$.

2. T decays *exponentially* with barrier width: for $\kappa w \gg 1$,

$$T \;\approx\; 16\,\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)\mathrm{e}^{-2\kappa w}. \tag{10}$$

3. In the above-barrier regime $E > V_0$, partial reflection still occurs due to the wave nature of ψ, with

$$T \;=\; \left[1 + \frac{V_0^2 \sin^2(k' w)}{4E(E - V_0)}\right]^{-1}, \quad k' = \sqrt{2(E - V_0)}. \tag{11}$$

   Resonant transmission ($T = 1$) occurs whenever $k' w = n\pi$.

3.2 Wave-packet dynamics

A Gaussian wave packet is a superposition of plane waves:

$$\psi(x, 0) \;=\; \int_{-\infty}^{+\infty} \tilde{\phi}(k)\,\mathrm{e}^{\mathrm{i}kx}\,\frac{dk}{2\pi}, \qquad \tilde{\phi}(k) \propto \mathrm{e}^{-(k-k_0)^2\sigma^2}\,\mathrm{e}^{-\mathrm{i}kx_0}. \tag{12}$$

Each component k evolves independently and interacts with the barrier with its own coefficient $T(k)$. After the scattering event, the total wave function splits into a *reflected* part travelling to the left and a *transmitted* part travelling to the right.

The integrated probabilities

$$R(t) = \int_{-\infty}^{x_{\text{bar}}} |\psi(x,t)|^2 \, dx, \tag{13}$$

$$T(t) = \int_{x_{\text{bar}}}^{+\infty} |\psi(x,t)|^2 \, dx \tag{14}$$

approach constant asymptotic values after the scattering is complete. These asymptotic values can be compared to the plane-wave formula (9) evaluated at $k = k_0$ — agreement is expected when the packet is narrow in k-space (large σ).

4 Numerical method

4.1 Spatial discretisation

The domain $[x_{\min}, x_{\max}]$ is divided into N uniformly spaced points with grid spacing $\Delta x = L/(N-1)$, where $L = x_{\max} - x_{\min}$. The wave function is represented as the vector $\boldsymbol{\psi}^n \in \mathbb{C}^N$ with components $\psi_i^n \approx \psi(x_i, n\Delta t)$.

The second derivative is replaced by the standard symmetric second-order finite-difference stencil:

$$\left.\frac{\partial^2 \psi}{\partial x^2}\right|_{x_i} \approx \frac{\psi_{i-1} - 2\psi_i + \psi_{i+1}}{\Delta x^2} + \mathcal{O}(\Delta x^2). \tag{15}$$

This approximates the kinetic part of $\hat{H}$ as a real symmetric tridiagonal matrix K with main diagonal $-2/\Delta x^2$ and off-diagonals $+1/\Delta x^2$ (multiplied by $-1/2$):

$$[H_{\text{disc}}]_{ij} = -\frac{1}{2\Delta x^2} \begin{pmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -2 \end{pmatrix}_{ij} + V_i \, \delta_{ij}. \tag{16}$$

4.2 Time discretisation: the Crank–Nicolson scheme

A naive explicit (forward-Euler) discretisation of the TDSE,

$$\boldsymbol{\psi}^{n+1} = \boldsymbol{\psi}^n - \mathrm{i}\Delta t \, H_{\text{disc}} \, \boldsymbol{\psi}^n, \tag{17}$$

is *conditionally stable*: the time step must satisfy $\Delta t \lesssim \Delta x^2$, which for $\Delta x = 0.01\,\text{bohr}$ means $\Delta t \lesssim 10^{-4}$ a.u. — two orders of magnitude smaller than the Crank–Nicolson step used here. Moreover, the explicit scheme does not conserve the norm exactly.

The **Crank–Nicolson scheme** [2] is an implicit second-order method that averages

the right-hand side between the current and the next time level:

$$\mathrm{i}\,\frac{\boldsymbol{\psi}^{n+1}-\boldsymbol{\psi}^{n}}{\Delta t} \;=\; \frac{H_{\mathrm{disc}}}{2}\left(\boldsymbol{\psi}^{n+1}+\boldsymbol{\psi}^{n}\right). \tag{18}$$

Rearranging (18):

$$\underbrace{\left(I+\frac{\mathrm{i}\Delta t}{2}\,H_{\mathrm{disc}}\right)}_{A}\boldsymbol{\psi}^{n+1} \;=\; \underbrace{\left(I-\frac{\mathrm{i}\Delta t}{2}\,H_{\mathrm{disc}}\right)}_{B}\boldsymbol{\psi}^{n}, \tag{19}$$

which is a linear system $A\,\boldsymbol{\psi}^{n+1}=\mathbf{b}^{n}$ with $\mathbf{b}^{n}=B\,\boldsymbol{\psi}^{n}$.

4.3 Properties of the Crank–Nicolson scheme

Unconditional stability. The eigenvalues of H_{disc} are real (it is Hermitian). For an eigenvalue λ, the amplification factor of the Crank–Nicolson scheme is

$$g \;=\; \frac{1-\mathrm{i}\Delta t\lambda/2}{1+\mathrm{i}\Delta t\lambda/2}, \tag{20}$$

which satisfies $|g|=1$ for *any* $\Delta t>0$. The scheme is therefore **unconditionally stable**: no time-step restriction is required for stability.

Norm (probability) conservation. The matrices A and B satisfy $B=A^{\dagger}$ (complex conjugate transpose). The discrete propagator $A^{-1}B$ is therefore *unitary*:

$$(A^{-1}B)^{\dagger}(A^{-1}B) \;=\; B^{\dagger}(A^{\dagger})^{-1}A^{-1}B \;=\; AA^{-1}(A^{\dagger})^{-1}B \;=\; I. \tag{21}$$

Consequently, $\|\boldsymbol{\psi}^{n+1}\|^2=\|\boldsymbol{\psi}^{n}\|^2$ at every step — the discrete norm is exactly conserved (up to floating-point rounding errors of order 10^{-14}).

Second-order accuracy. The local truncation error of (18) is $\mathcal{O}(\Delta t^2)$ in time and $\mathcal{O}(\Delta x^2)$ in space, so the scheme is globally second-order in both variables.

4.4 Matrix structure

The matrices A and B are *complex tridiagonal*. Defining the parameter

$$r \;=\; \frac{\Delta t}{4\Delta x^2}, \tag{22}$$

their non-zero elements are (for $1 \le i \le N-2$):

$$A_{i,i} = 1 + \mathrm{i}\frac{\Delta t}{2}\left(\frac{1}{\Delta x^2} + V_i\right), \quad A_{i,i\pm1} = -\mathrm{i}r, \tag{23}$$

$$B_{i,i} = 1 - \mathrm{i}\frac{\Delta t}{2}\left(\frac{1}{\Delta x^2} + V_i\right), \quad B_{i,i\pm1} = +\mathrm{i}r. \tag{24}$$

The first and last rows enforce Dirichlet boundary conditions $\psi_0 = \psi_{N-1} = 0$: both rows are set to the identity row $(1, 0, \ldots)$ in A and B.

Because $V(x)$ is static, A and B are **built once** before the time loop and reused at every step.

4.5 The Thomas algorithm

At each time step, the right-hand side is evaluated as a matrix-vector product $\mathbf{b}^n = B\,\boldsymbol{\psi}^n$ (one pass over the grid, $\mathcal{O}(N)$ operations), and the resulting tridiagonal system $A\,\boldsymbol{\psi}^{n+1} = \mathbf{b}^n$ is solved with the **Thomas algorithm** [3].

The Thomas algorithm is Gaussian elimination specialised for tridiagonal systems. For a system with lower diagonal l_i, main diagonal d_i, upper diagonal u_i, and right-hand side b_i:

Forward sweep (elimination of l_i, from $i = 1$ to $N-1$):

$$w_i = l_i/d_{i-1}, \tag{25}$$

$$d_i \leftarrow d_i - w_i\,u_{i-1}, \tag{26}$$

$$b_i \leftarrow b_i - w_i\,b_{i-1}. \tag{27}$$

Back substitution (from $i = N-1$ down to 0):

$$x_{N-1} = b_{N-1}/d_{N-1}, \tag{28}$$

$$x_i = (b_i - u_i\,x_{i+1})/d_i. \tag{29}$$

The total cost is $\mathcal{O}(N)$ in both time and memory — optimal for a tridiagonal system. The algorithm is numerically stable for diagonally dominant matrices, which the Crank–Nicolson matrix A always satisfies (the main diagonal dominates because $|A_{ii}| > 2|A_{i,i\pm1}|$ whenever $\Delta t/\Delta x^2 < 2$, easily satisfied in practice).

This is the same algorithm implemented by LAPACK's routine `zgtsv`, here realised explicitly in complex arithmetic without any external library dependency.

4.6 Boundary conditions

Dirichlet boundary conditions $\psi(x_{\min}, t) = \psi(x_{\max}, t) = 0$ are imposed. Physically, the walls act as infinite potential barriers. They introduce artificial reflections if the wave packet reaches the boundary, so the domain length L must be chosen large enough that the packet remains far from the walls throughout the simulation.

A practical criterion is $x_{\min} \ll x_0 - k_0 T_{\text{tot}}$ and $x_{\max} \gg x_0 + k_0 T_{\text{tot}}$, where $k_0 T_{\text{tot}}$ is the approximate distance travelled by the centre of the packet.

4.7 Algorithm summary

The complete time-stepping algorithm is:

1. Read parameters and build grid $\{x_i\}$ and potential $\{V_i\}$.
2. Initialise $\boldsymbol{\psi}^0$ as the normalised Gaussian (6).
3. Build the tridiagonal matrices A and B (once).
4. **For** $n = 0, 1, \ldots, M-1$:
   (a) Compute $\mathbf{b}^n = B\,\boldsymbol{\psi}^n$ (matrix-vector product, $\mathcal{O}(N)$).
   (b) Solve $A\,\boldsymbol{\psi}^{n+1} = \mathbf{b}^n$ with Thomas algorithm ($\mathcal{O}(N)$).
   (c) Enforce $\psi_0^{n+1} = \psi_{N-1}^{n+1} = 0$.
   (d) Write snapshot and norm to file (if snapshot step).

Total cost per time step: $\mathcal{O}(N)$. Total cost: $\mathcal{O}(MN)$.

5 Analysis of the simulation output

This section describes the expected physical behaviour and the corresponding signatures in each of the three plots produced by the Python analysis scripts.

The default simulation parameters are:

Parameter	Value	Role
L	20 bohr	domain length
Δx	0.01 bohr	spatial step
Δt	0.001 a.u.	time step
T_{tot}	2.0 a.u.	total time
x_0	-5.0 bohr	initial packet centre
σ	0.5 bohr	packet width
k_0	5.0 bohr^{-1}	central wave vector
x_{bar}	0.0 bohr	barrier left edge
w	0.5 bohr	barrier width
V_0	20.0 E_h	barrier height

The mean kinetic energy of the packet is $E = k_0^2/2 = 12.5\,E_h$, while the barrier height is $V_0 = 20.0\,E_h$, so $E < V_0$: the simulation is in the **tunnelling regime**. The evanescent decay constant inside the barrier is $\kappa = \sqrt{2(V_0 - E)} \approx 3.87\,\text{bohr}^{-1}$, giving an approximate transmission probability from (9):

$$T_{\text{analytic}} \approx \left[1 + \frac{20^2 \sinh^2(3.87 \times 0.5)}{4 \times 12.5 \times 7.5}\right]^{-1} \approx 0.04. \tag{30}$$

About 4% of the probability is expected to tunnel through the barrier, with the remaining 96% reflected.

5.1 Snapshot plot (`snapshots.png`)

What is shown. Five panels display $|\psi(x,t)|^2$ at equally-spaced times $t = 0,\ T/4,\ T/2,\ 3T/4,\ T$. The potential barrier is shown as a shaded region scaled to the maximum probability density for visual reference.

Expected behaviour.

- $t = 0$: A single Gaussian peak centred at $x_0 = -5$ bohr, well to the left of the barrier. Its width is $\sigma = 0.5$ bohr.
- **Propagation phase** ($t \approx T/4$): The packet moves rightward at group velocity $v_g = k_0 = 5$ bohr/a.u. It gradually broadens due to dispersion: because different k components travel at slightly different speeds $v(k) = k$, the packet spreads as $\sigma(t) \approx \sqrt{\sigma^2 + t^2/(4\sigma^2)}$.
- **Scattering** ($t \approx T/2$): The packet reaches the barrier. The probability density is temporarily compressed and distorted — the leading edge decelerates while the tail catches up. A small transmitted fraction leaks through the right side of the barrier.

- **Post-scattering ($t \approx 3T/4$ and T)**: Two distinct peaks are visible. A large reflected packet travels leftward; a small transmitted packet continues rightward. The relative areas under the two peaks reflect the ratio $R : T \approx 96 : 4$.

What to look for. The total area under the curve (norm) must remain constant across all panels. Any visible growth or decay signals a numerical problem.

5.2 Quantitative analysis plot (`analysis.png`)

5.2.1 Panel A: Norm conservation

What is shown. The discrete norm $\|\boldsymbol{\psi}^n\|_\Delta^2 x = \Delta x \sum_i |\psi_i^n|^2$ as a function of time, compared against the exact value of 1.

Expected behaviour. Because the Crank–Nicolson propagator is unitary (Section 4.3), the norm deviation should be $|\|\psi^n\|^2 - 1| \lesssim 10^{-12}$ throughout — essentially machine precision. This is fundamentally different from explicit schemes, which dissipate or amplify the norm.

What to look for. A flat line at 1 confirms the unitarity of the numerical propagator. Any monotonic drift would indicate an implementation bug; oscillatory drift would suggest stability issues.

5.2.2 Panel B: Transmission and reflection coefficients

What is shown. The time evolution of

$$R(t) = \Delta x \sum_{i:\, x_i < x_{\text{bar}}} |\psi_i|^2, \qquad T(t) = \Delta x \sum_{i:\, x_i \geq x_{\text{bar}}} |\psi_i|^2, \tag{31}$$

together with their sum $R(t) + T(t)$.

Expected behaviour. Initially $R \approx 1$ and $T \approx 0$ (entire packet to the left of the barrier). As the packet reaches and interacts with the barrier, R decreases and T increases. After the scattering is complete, both coefficients reach their asymptotic values: $T_\infty \approx 0.04$ and $R_\infty \approx 0.96$ from (30).

The sum $R(t) + T(t)$ should remain equal to 1 at all times (it is just another way of writing the norm conservation), providing a self-consistency check. During the scattering phase, $R + T$ may appear to dip very slightly if the wave function is spread *across* the barrier boundary, but this is a boundary effect of the integration region, not a physical violation.

What to look for. The asymptotic value of T_∞ can be directly compared to the analytic plane-wave result (30). Discrepancies arise because the Gaussian packet has a finite momentum spread; the wave-packet result converges to the plane-wave formula in the limit $\sigma \to \infty$ (narrow k-distribution).

5.2.3 Panel C: Mean position $\langle x \rangle(t)$

What is shown. The expectation value of position,

$$\langle x \rangle(t) \;=\; \Delta x \sum_i x_i \, |\psi_i(t)|^2, \tag{32}$$

as a function of time.

Expected behaviour.

- **Before the barrier**: $\langle x \rangle$ increases linearly with slope $v_g = k_0 = 5\,\text{bohr/a.u.}$, reflecting the classical motion of the packet centre.
- **During scattering**: the slope decreases as the packet interacts with the barrier. Classically, the particle would be reflected instantaneously; quantum mechanically, the interaction is spread over a finite time (related to the *Wigner delay time*).
- **After scattering**: $\langle x \rangle$ is the probability-weighted average of the reflected and transmitted centres. Since $R \gg T$, the mean position moves leftward (the reflected packet dominates). The slope of the asymptotic linear regime will be negative, approximately $-k_0 R_\infty$.

What to look for. The transition from positive to negative slope signals the moment of scattering. The asymptotic slope encodes the relative weight of reflected vs. transmitted components.

5.2.4 Panel D: Packet width $\sigma(t)$

What is shown. The root-mean-square width of the packet,

$$\sigma(t) \;=\; \sqrt{\langle x^2 \rangle - \langle x \rangle^2}, \quad \langle x^2 \rangle = \Delta x \sum_i x_i^2 \, |\psi_i|^2. \tag{33}$$

Expected behaviour.

- **Free propagation**: A Gaussian wave packet in free space spreads as

$$\sigma(t) \;=\; \sigma_0 \sqrt{1 + \left(\frac{t}{2\sigma_0^2} \right)^2}, \tag{34}$$

which for $t \gg 2\sigma_0^2 = 0.5\,\text{a.u.}$ becomes linear: $\sigma(t) \approx t/(2\sigma_0)$.

- **After scattering**: The width grows faster than the free-space result. The physical reason is that the wave function splits into two spatially separated components (reflected and transmitted), and the variance $\langle x^2\rangle - \langle x\rangle^2$ receives a large contribution from the *distance between the two peaks*, not just from the width of each individual peak. This is a purely quantum interference effect.

What to look for. A clear change of slope at the time of scattering: the growth rate of $\sigma(t)$ increases sharply when the packet splits.

5.3 Animation (`evolution.mp4`)

The animation simultaneously shows $|\psi(x,t)|^2$ (solid blue) and $\mathrm{Re}[\psi(x,t)]$ (dashed orange), which reveals the oscillatory phase structure of the wave function.

Expected behaviour.

- The real part oscillates at spatial frequency k_0, visible as rapid oscillations inside the packet envelope.
- As the packet approaches the barrier, the oscillations in the barrier region are exponentially suppressed (evanescent wave).
- After the interaction, the reflected packet shows oscillations travelling in the $-x$ direction (decreasing phase with time), while the tiny transmitted packet oscillates in the $+x$ direction.
- Interference between the incoming and reflected components is visible as a standing-wave pattern to the left of the barrier during the scattering phase.

6 Numerical parameters: stability and accuracy

6.1 Grid resolution requirements

For the simulation to faithfully represent the wave packet, the grid must resolve the shortest physical length scales:

- **Spatial**: the de Broglie wavelength $\lambda = 2\pi/k_0$ must be resolved with several grid points. For $k_0 = 5$, $\lambda \approx 1.26\,\text{bohr}$, and $\Delta x = 0.01\,\text{bohr}$ gives ≈ 126 points per wavelength — more than sufficient.

- **Inside the barrier**: the evanescent decay length is $1/\kappa \approx 0.26\,\text{bohr}$ for the default parameters. With $\Delta x = 0.01$, there are ≈ 26 points per decay length, which is adequate.

- **Temporal**: the Crank–Nicolson scheme is unconditionally stable, so Δt is constrained only by *accuracy*. A practical criterion is $\Delta t \ll 1/E_{\max}$, where $E_{\max} \approx k_{\max}^2/2$ is the maximum energy in the spectrum. For $k_{\max} \approx k_0 + 3/(2\sigma) \approx 8$, $E_{\max} \approx 32\,\text{a.u.}$, giving $\Delta t \ll 0.03\,\text{a.u.}$ The default $\Delta t = 0.001\,\text{a.u.}$ satisfies this easily.

6.2 Courant number

Although the Crank–Nicolson scheme is unconditionally stable, the effective Courant number $\nu = k_0 \Delta t/\Delta x = 5 \times 0.001/0.01 = 0.5$ gives an indication of the accuracy of the phase velocity. Values $\nu \lesssim 1$ ensure that the numerical group velocity closely matches the analytical one.

7 Conclusions

The Crank–Nicolson scheme provides a robust, second-order, norm-conserving method for propagating the TDSE. Its implicit nature eliminates the time-step restriction that affects explicit schemes, and the tridiagonal structure of the resulting linear system makes each step $\mathcal{O}(N)$ via the Thomas algorithm.

The simulation reproduces the key quantum features of barrier scattering: wave-packet spreading, tunnelling through a classically forbidden region, and the splitting of the packet into reflected and transmitted components. The quantitative comparison of the numerically computed transmission coefficient with the analytic plane-wave formula provides a direct validation of both the physical model and the numerical implementation.

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A The quantum state space and Dirac formalism

A.1 Hilbert space structure

The rigorous mathematical setting for quantum mechanics is a *complex separable Hilbert space* $\mathcal{H}$. For a one-dimensional spinless particle, the natural choice is $\mathcal{H} = L^2(\mathbb{R})$: the space of square-integrable complex functions on $\mathbb{R}$, equipped with the inner product

$$\langle f, g \rangle \;=\; \int_{-\infty}^{+\infty} f^*(x)\, g(x)\, dx. \tag{35}$$

A physical state corresponds to a *unit vector* (ray) in $\mathcal{H}$, i.e. $\|\psi\| = 1$.

In Dirac notation, states are *kets* $|\psi\rangle \in \mathcal{H}$ and their duals are *bras* $\langle\psi| \in \mathcal{H}^*$. The inner product is $\langle\phi|\psi\rangle$, and the norm is $\|\psi\|^2 = \langle\psi|\psi\rangle$.

A.2 Observables as self-adjoint operators

Every physical observable A corresponds to a *self-adjoint* (Hermitian) operator $\hat{A} : \mathcal{H} \to \mathcal{H}$, defined by $\hat{A}^\dagger = \hat{A}$. Self-adjointness guarantees:

- All eigenvalues are real (measurable quantities must be real).
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- The eigenvectors form a complete orthonormal basis of $\mathcal{H}$ (spectral theorem).

For continuous spectra (position, momentum), the eigenstates are distributions rather than true L^2 functions. The position eigenstates $|x'\rangle$ satisfy

$$\hat{x}\,|x'\rangle = x'\,|x'\rangle\,, \qquad \langle x|x'\rangle = \delta(x - x'), \qquad \int_{-\infty}^{+\infty} |x'\rangle\,\langle x'|\; dx' = \hat{I}, \tag{36}$$

and the wave function is the projection $\psi(x,t) = \langle x|\psi(t)\rangle$.

A.3 The canonical commutation relation

The position and momentum operators satisfy

$$[\hat{x},\, \hat{p}] \;=\; \mathrm{i}\hbar\, \hat{I}. \tag{37}$$

In the position representation, $\hat{p} = -\mathrm{i}\hbar\, \partial_x$. The commutation relation (37) is the algebraic root of the Heisenberg uncertainty principle: for any state $|\psi\rangle$,

$$\Delta x\, \Delta p \;\geq\; \frac{\hbar}{2}, \tag{38}$$

where $(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$. The Gaussian wave packet (6) saturates this bound, making it the *coherent state* of the Weyl–Heisenberg algebra.

A.4 Time evolution operator and unitarity

The formal solution of the TDSE is

$$|\psi(t)\rangle \;=\; \hat{U}(t,t_0)\;|\psi(t_0)\rangle\,, \qquad \hat{U}(t,t_0) = \exp\left[-\frac{\mathrm{i}}{\hbar}\hat{H}(t-t_0)\right], \tag{39}$$

where the last expression holds for time-independent $\hat{H}$. The operator $\hat{U}$ is *unitary*: $\hat{U}^\dagger\hat{U} = \hat{I}$, which is the operator statement of probability conservation. The Crank–Nicolson propagator $A^{-1}B$ is a finite-dimensional unitary approximation to $\hat{U}(\Delta t)$.

B Transfer matrix method and exact tunnelling formulae

B.1 Piecewise solution of the stationary Schrödinger equation

For a static potential the TDSE separates via $\psi(x,t) = \phi(x)\,\mathrm{e}^{-\mathrm{i}Et/\hbar}$, yielding the *time-independent Schrödinger equation* (TISE):

$$-\frac{1}{2}\phi''(x) + V(x)\phi(x) \;=\; E\,\phi(x). \tag{40}$$

For the rectangular barrier (8), the TISE has piecewise constant coefficients. In each region the general solution is:

$$\text{Region I } (x < x_{\text{bar}}): \quad \phi(x) = A\,\mathrm{e}^{\mathrm{i}kx} + B\,\mathrm{e}^{-\mathrm{i}kx}, \quad k = \sqrt{2E}, \tag{41}$$

$$\text{Region II } (x_{\text{bar}} < x < x_{\text{bar}} + w): \quad \phi(x) = C\,\mathrm{e}^{\kappa x} + D\,\mathrm{e}^{-\kappa x}, \quad \kappa = \sqrt{2(V_0 - E)}, \quad (E < V_0), \tag{42}$$

$$\text{Region III } (x > x_{\text{bar}} + w): \quad \phi(x) = F\,\mathrm{e}^{\mathrm{i}kx}, \tag{43}$$

where $F\mathrm{e}^{\mathrm{i}kx}$ is the transmitted wave (no left-moving component in region III for a right-incident wave).

B.2 The transfer matrix

At each interface, continuity of ϕ and ϕ' provides two matching conditions. These can be compactly encoded in a 2×2 *transfer matrix* M such that

$$\begin{pmatrix} A \\ B \end{pmatrix} = M \begin{pmatrix} F \\ 0 \end{pmatrix}. \tag{44}$$

For the rectangular barrier, $M = M_{\mathrm{I}\to\mathrm{II}} \cdot M_{\mathrm{II}\to\mathrm{III}}$, where each interface matrix is

$$M_{j\to k} = \frac{1}{2q_k} \begin{pmatrix} q_k + q_j & q_k - q_j \\ q_k - q_j & q_k + q_j \end{pmatrix}, \tag{45}$$

with q_j being the (possibly imaginary) wave vector in region j. The transmission amplitude is $t = 1/M_{11}$ and

$$T = |t|^2 = \frac{1}{|M_{11}|^2}, \tag{46}$$

which, after explicit computation, reproduces (9). The transfer matrix formalism generalises straightforwardly to arbitrary piecewise-constant potentials.

B.3 WKB approximation

For a *smooth* potential barrier, the exact transfer matrix is not available in closed form. The Wentzel–Kramers–Brillouin (WKB) approximation replaces the exact evanescent solution inside the barrier with a locally-plane-wave ansatz:

$$\phi(x) \approx \frac{C}{\sqrt{\kappa(x)}} \exp\left[-\int_{x_1}^{x} \kappa(x')\, dx'\right], \quad \kappa(x) = \sqrt{2(V(x) - E)}, \tag{47}$$

where x_1, x_2 are the classical turning points ($V(x_{1,2}) = E$). The WKB transmission coefficient is

$$T_{\mathrm{WKB}} \approx \exp\left[-2\int_{x_1}^{x_2} \kappa(x)\, dx\right]. \tag{48}$$

For the rectangular barrier, $\kappa(x) = \kappa = \text{const}$ and (48) gives $T \approx \mathrm{e}^{-2\kappa w}$, consistent with the large-κw limit of (9).

C Dispersion relation and free-packet dynamics

C.1 Dispersion relation

For a free particle ($V = 0$), substituting a plane wave $\psi = \mathrm{e}^{\mathrm{i}(kx-\omega t)}$ into the TDSE gives the *dispersion relation*

$$\omega(k) = \frac{k^2}{2} \quad \text{(atomic units)}. \tag{49}$$

This is a *quadratic* dispersion, unlike the linear dispersion of photons. Two velocities characterise wave-packet propagation:

$$v_{\text{phase}} = \frac{\omega}{k} = \frac{k}{2}, \tag{50}$$

$$v_{\text{group}} = \frac{d\omega}{dk} = k. \tag{51}$$

The group velocity $v_g = k_0$ is the velocity of the packet's centre of mass and equals the classical velocity $p/m = k_0$ in atomic units.

C.2 Exact free-propagation of the Gaussian packet

A Gaussian wave packet in free space can be propagated exactly using the Fourier convolution theorem. Starting from (6), its Fourier transform is

$$\tilde{\phi}(k,0) \;=\; (2\pi\sigma^2)^{1/4}\,\sigma\,\mathrm{e}^{-(k-k_0)^2\sigma^2}\,\mathrm{e}^{-\mathrm{i}kx_0}. \tag{52}$$

At time t, each mode acquires a phase $\mathrm{e}^{-\mathrm{i}\omega(k)t}$:

$$\tilde{\phi}(k,t) \;=\; \tilde{\phi}(k,0)\,\mathrm{e}^{-\mathrm{i}k^2t/2}. \tag{53}$$

Transforming back gives

$$\psi(x,t) \;=\; \left(\frac{1}{2\pi\sigma_t^2}\right)^{1/4} \exp\left[-\frac{(x-x_0-k_0t)^2}{4\sigma_t^2}\right] \exp\left[\mathrm{i}k_0\left(x-\frac{k_0t}{2}\right)\right] \exp[\mathrm{i}\phi_t]\,, \tag{54}$$

where the *complex width* is

$$\sigma_t \;=\; \sigma\sqrt{1+\mathrm{i}t/(2\sigma^2)}, \tag{55}$$

and $\phi_t = -\frac{1}{2}\arg(\sigma_t/\sigma)$ is a global phase. The probability density is still Gaussian:

$$|\psi(x,t)|^2 \;=\; \frac{1}{\sqrt{2\pi}|\sigma_t|}\exp\left[-\frac{(x-x_0-k_0t)^2}{2|\sigma_t|^2}\right], \tag{56}$$

with real width $|\sigma_t| = \sigma\sqrt{1+t^2/(4\sigma^4)}$, which is exactly equation (34).

The spreading rate $d|\sigma_t|/dt \to 1/(2\sigma)$ for large t is inversely proportional to the initial width: a narrow packet (small σ, large momentum spread) spreads faster.

D Operator splitting and the Cayley form

D.1 The exact propagator as a matrix exponential

The formal solution of the TDSE over one time step is

$$\boldsymbol{\psi}^{n+1} \;=\; \mathrm{e}^{-\mathrm{i}\Delta t H_{\mathrm{disc}}}\,\boldsymbol{\psi}^{n}. \tag{57}$$

Direct computation of the matrix exponential costs $\mathcal{O}(N^3)$ — prohibitive for large N. Several efficient approximations exist.

D.2 Padé approximants and the Cayley form

The Crank–Nicolson scheme is equivalent to the *first-order Padé approximant* of the matrix exponential:

$$\mathrm{e}^{-\mathrm{i}\Delta t H} \;\approx\; \frac{I - \mathrm{i}\Delta t H/2}{I + \mathrm{i}\Delta t H/2} \;=\; A^{-1}B. \tag{58}$$

This is also known as the *Cayley form* of the propagator. It is *exactly unitary* regardless of Δt, because the numerator and denominator are complex conjugates of each other (for Hermitian H).

Higher-order Padé approximants $[m/m]$ exist and achieve $\mathcal{O}(\Delta t^{2m})$ accuracy while remaining unitary, but they require solving m tridiagonal systems per step and are rarely used in practice.

D.3 Strang splitting

For Hamiltonians of the form $H = T + V$ where T is the kinetic operator and V is diagonal in position space, the *split-operator* (Strang splitting) method approximates:

$$\mathrm{e}^{-\mathrm{i}\Delta t(T+V)} \;\approx\; \mathrm{e}^{-\mathrm{i}\Delta t V/2}\,\mathrm{e}^{-\mathrm{i}\Delta t T}\,\mathrm{e}^{-\mathrm{i}\Delta t V/2} + \mathcal{O}(\Delta t^3). \tag{59}$$

Each factor can be applied efficiently: $\mathrm{e}^{-\mathrm{i}\Delta t V/2}$ is a pointwise multiplication in position space (diagonal), while $\mathrm{e}^{-\mathrm{i}\Delta t T}$ is a multiplication in momentum (Fourier) space. Using the Fast Fourier Transform (FFT), each step costs $\mathcal{O}(N \log N)$, compared to $\mathcal{O}(N)$ for the Crank–Nicolson Thomas algorithm. However, the split-operator method is second-order accurate and exactly unitary only in the limit of exact arithmetic, and requires global FFTs which are harder to parallelise.

E Scattering theory: S-matrix and Wigner delay time

E.1 The scattering matrix

For a potential that vanishes at $x \to \pm\infty$, scattering theory describes the asymptotic relation between incoming and outgoing plane waves. In 1D, the *S-matrix* is a 2×2 unitary matrix:

$$S = \begin{pmatrix} t & r' \\ r & t' \end{pmatrix}, \tag{60}$$

where t, t' are transmission amplitudes (right-incident and left-incident) and r, r' are reflection amplitudes. Unitarity $S^\dagger S = I$ implies $|t|^2 + |r|^2 = 1$, i.e. $T + R = 1$.

For a symmetric potential (even $V(x)$), $t = t'$ and $|r| = |r'|$. The rectangular barrier is symmetric about its centre, so both transmission amplitudes are equal.

E.2 Phase shifts and time delays

The transmission amplitude can be written as $t = \sqrt{T}\,\mathrm{e}^{\mathrm{i}\varphi_t(k)}$, where φ_t is the transmission phase. The *Wigner–Smith delay time* is defined as

$$\tau(k) = \frac{d\varphi_t}{d\omega} = \frac{1}{v_g}\frac{d\varphi_t}{dk}, \tag{61}$$

and represents the extra time spent by the transmitted wave packet near the barrier compared to free propagation [10]. For the rectangular barrier in the tunnelling regime, $\tau > 0$ (the packet is delayed), and as $V_0 \to \infty$ the delay saturates to $\tau_{\max} = w/v_g$ — the so-called *Hartman effect*, where the delay becomes independent of barrier width for opaque barriers.

In the simulation, the delay manifests as a reduction in the slope of $\langle x \rangle(t)$ during the scattering event (Panel C of `analysis.png`).

E.3 The optical theorem in 1D

The optical theorem in 1D relates the forward scattering amplitude to the total cross-section. In one dimension, "cross-section" is replaced by the *total reflectance*, and the theorem states:

$$\mathrm{Im}(r) + \frac{k}{2}(T - 1) = 0, \tag{62}$$

which is a direct consequence of S-matrix unitarity and provides an additional consistency check for numerical scattering calculations.

F Von Neumann stability analysis

F.1 Von Neumann analysis for the TDSE

The stability of finite-difference schemes for linear PDEs can be analysed rigorously via the von Neumann (Fourier) method. Consider a uniform grid and a single Fourier mode $\psi_j^n = g^n \, \mathrm{e}^{\mathrm{i}j\xi\Delta x}$, where ξ is the wave vector and g is the amplification factor. Stability requires $|g| \leq 1$ for all ξ.

Forward Euler (explicit): Substituting into $\psi_j^{n+1} = \psi_j^n - \mathrm{i}\Delta t H_{\mathrm{disc}} \psi_j^n$ gives

$$g \;=\; 1 - \mathrm{i}\Delta t \, \lambda(\xi), \quad \lambda(\xi) = \frac{2(1-\cos(\xi\Delta x))}{\Delta x^2} + V \in \mathbb{R}. \tag{63}$$

Then $|g|^2 = 1 + \Delta t^2 \lambda^2 > 1$ for $\lambda \neq 0$: the scheme is *always unstable* for the Schrödinger equation.

Crank–Nicolson: The amplification factor is

$$g \;=\; \frac{1 - \mathrm{i}\Delta t \lambda/2}{1 + \mathrm{i}\Delta t \lambda/2}. \tag{64}$$

Since $\lambda \in \mathbb{R}$, we have $|g|^2 = 1$ identically for all ξ and all $\Delta t > 0$: the scheme is *unconditionally stable* and, moreover, *exactly norm-preserving* at the level of the amplification factor.

Implicit mid-point rule (Crank–Nicolson is a special case): More generally, the θ-method

$$\mathrm{i}\frac{\psi^{n+1} - \psi^n}{\Delta t} \;=\; (1-\theta) H \psi^n + \theta H \psi^{n+1} \tag{65}$$

is stable for $\theta \geq 1/2$. The Crank–Nicolson scheme corresponds to $\theta = 1/2$ and is the only choice in this family that is both stable and second-order accurate.

F.2 Numerical dispersion

Although the Crank–Nicolson scheme is unconditionally stable, it introduces a *numerical dispersion error*: the discrete phase velocity $\tilde{\omega}(\xi)$ differs from the exact $\omega(\xi) = \xi^2/2$. From the amplification factor,

$$\tilde{\omega}(\xi) \;=\; \frac{2}{\Delta t} \arctan\!\left(\frac{\Delta t \, \lambda(\xi)}{2}\right), \quad \lambda(\xi) = \frac{2(1-\cos\xi\Delta x)}{\Delta x^2}. \tag{66}$$

Expanding for small $\xi\Delta x$ and Δt:

$$\tilde{\omega}(\xi) \;=\; \frac{\xi^2}{2}\left[1 - \frac{(\xi\Delta x)^2}{12} - \frac{\Delta t^2 \xi^4}{24} + \dots\right], \tag{67}$$

showing $\mathcal{O}(\Delta x^2, \Delta t^2)$ dispersion errors, consistent with the second-order accuracy of the scheme. High-k components of the wave packet travel at a slightly wrong speed — this is the physical origin of the *numerical spreading* that adds to the physical quantum spreading.

G Ehrenfest's theorem and quantum-classical correspondence

G.1 Statement and derivation

Ehrenfest's theorem establishes the connection between quantum expectation values and classical equations of motion. For any observable $\hat{A}$, the Heisenberg equation of motion is:

$$\frac{d}{dt}\langle \hat{A}\rangle \;=\; \frac{1}{\mathrm{i}\hbar}\langle[\hat{A}, \hat{H}]\rangle + \left\langle \frac{\partial \hat{A}}{\partial t}\right\rangle. \tag{68}$$

Applying this to $\hat{x}$ and $\hat{p}$ with $\hat{H} = \hat{p}^2/2m + V(\hat{x})$:

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{\langle\hat{p}\rangle}{m}, \tag{69}$$

$$\frac{d\langle\hat{p}\rangle}{dt} = -\left\langle \frac{\partial V}{\partial x}\right\rangle. \tag{70}$$

These are the quantum analogues of Hamilton's equations. Note the crucial difference from the classical result: (70) involves $\langle V'(\hat{x})\rangle$, not $V'(\langle\hat{x}\rangle)$. The two coincide only if V is at most quadratic, or if the wave packet is so narrow that $\psi(x,t) \approx \delta(x - x_{\mathrm{cl}}(t))$.

G.2 Implications for the simulation

In the early propagation phase, the wave packet is narrow compared to the scale on which V varies (the barrier), so Ehrenfest's theorem predicts nearly classical motion: $\langle x\rangle(t) \approx x_0 + k_0 t$. This is exactly the linear regime visible in Panel C before the scattering event.

When the packet interacts with the barrier, $\langle V'\rangle \neq V'(\langle x\rangle)$, the classical approximation breaks down, and quantum effects (tunnelling, interference) dominate. After the scattering, the wave function consists of two well-separated packets, and $\langle x\rangle$ becomes a weighted average that has no classical counterpart.

G.3 Energy conservation

From Ehrenfest's theorem, the mean energy $\langle \hat{H} \rangle$ is conserved:

$$\frac{d}{dt}\langle \hat{H} \rangle \;=\; \frac{1}{\mathrm{i}\hbar}\langle [\hat{H}, \hat{H}] \rangle = 0. \tag{71}$$

In the numerical simulation, energy conservation is not enforced explicitly by the Crank–Nicolson scheme (only norm is exactly conserved). Nevertheless, the second-order accuracy of the scheme ensures that the energy drift is $\mathcal{O}(\Delta t^2)$ per step.

H Absorbing boundary conditions and perfectly matched layers

H.1 The problem with Dirichlet boundaries

Dirichlet boundary conditions $\psi(x_{\min}) = \psi(x_{\max}) = 0$ mimic infinitely hard walls. They are simple to implement but introduce *spurious reflections* as soon as the wave packet reaches the boundary. The domain must therefore be large enough that no part of ψ reaches the walls during the simulation — a requirement that can make the problem expensive for long times or fast packets.

H.2 Absorbing boundary conditions

A more efficient alternative is to modify the Hamiltonian near the boundaries by adding a negative imaginary potential (absorption):

$$V(x) \;\to\; V(x) - \mathrm{i}\,\Gamma(x), \quad \Gamma(x) \geq 0, \tag{72}$$

where $\Gamma(x)$ is large only near the domain edges. This makes the Hamiltonian *non-Hermitian* in the absorbing region, causing the norm to decay as probability is absorbed. The reflected numerical wave is exponentially suppressed.

A standard choice is the cosine ramp:

$$\Gamma(x) \;=\; \Gamma_0 \sin^2\!\left(\frac{\pi}{2} \cdot \frac{|x| - x_{\mathrm{abs}}}{L_{\mathrm{abs}}}\right) \quad \text{for } |x| > x_{\mathrm{abs}}, \tag{73}$$

where L_{abs} and Γ_0 are the absorption layer width and strength.

H.3 Perfectly matched layers

The gold standard for absorbing boundaries is the *perfectly matched layer* (PML), originally developed for Maxwell's equations [5] and later adapted to the TDSE. A PML is

constructed by analytically continuing the spatial coordinate into the complex plane:

$$x \;\to\; \tilde{x}(x) \;=\; x + \mathrm{i}\int_0^x \sigma(x')\,dx', \tag{74}$$

where $\sigma(x') \geq 0$ is the PML absorption profile. This transformation preserves the form of the Schrödinger equation in the PML region while making all outgoing waves evanescent — and crucially, *without creating any reflection at the PML interface*, regardless of the angle or frequency of the incoming wave.

For the 1D TDSE, the PML modification is equivalent to introducing a complex coordinate stretching factor $s(x)$:

$$\hat{H}_{\mathrm{PML}} \;=\; -\frac{1}{2s(x)}\frac{\partial}{\partial x}\frac{1}{s(x)}\frac{\partial}{\partial x} + V(x), \quad s(x) = 1 + \mathrm{i}\sigma(x)/\omega. \tag{75}$$

The present simulation uses simple Dirichlet BCs, which are sufficient for the default parameters. Implementing absorbing BCs would allow simulations on much smaller domains or over much longer times.